

Exercise 3: Stored Procedures

Creating Table:

```
CREATE TABLE Accounts (  
    AccountID INT PRIMARY KEY,  
    CustomerName VARCHAR2(100),  
    AccountType VARCHAR2(20),  
    Balance NUMBER(15, 2)  
);
```

```
CREATE TABLE Employees (  
    EmployeeID INT PRIMARY KEY,  
    Name VARCHAR2(100),  
    Department VARCHAR2(50),  
    Salary NUMBER(15, 2)  
);
```

```
INSERT INTO Accounts VALUES (1, 'Alice', 'Savings', 5000);  
INSERT INTO Accounts VALUES (2, 'Bob', 'Savings', 2000);  
INSERT INTO Accounts VALUES (3, 'Charlie', 'Checking', 1000);
```

```
INSERT INTO Employees VALUES (101, 'John Doe', 'Sales', 50000);  
INSERT INTO Employees VALUES (102, 'Jane Smith', 'Sales', 60000);  
INSERT INTO Employees VALUES (103, 'Mike Ross', 'HR', 45000);
```

```
COMMIT;
```

Scenario 1:

```
CREATE OR REPLACE PROCEDURE ProcessMonthlyInterest AS  
BEGIN  
    UPDATE Accounts  
    SET Balance = Balance * 1.01  
    WHERE AccountType = 'Savings';  
  
    COMMIT;  
    DBMS_OUTPUT.PUT_LINE('Monthly interest applied to all Savings accounts.');
```

END;
/

Scenario 2:

```
CREATE OR REPLACE PROCEDURE UpdateEmployeeBonus (  
    p_dept IN VARCHAR2,  
    p_bonus_percentage IN NUMBER  
) AS  
BEGIN  
    UPDATE Employees  
    SET Salary = Salary + (Salary * (p_bonus_percentage / 100))  
    WHERE Department = p_dept;  
  
    COMMIT;  
    DBMS_OUTPUT.PUT_LINE('Salaries updated for Department: ' || p_dept);  
END;  
/
```

Scenario 3:

```
CREATE OR REPLACE PROCEDURE TransferFunds (  
    p_from_account IN INT,  
    p_to_account IN INT,  
    p_amount IN NUMBER  
) AS  
    v_balance NUMBER;  
BEGIN  
  
    SELECT Balance INTO v_balance FROM Accounts WHERE AccountID = p_from_account;  
  
    IF v_balance < p_amount THEN  
        DBMS_OUTPUT.PUT_LINE('Error: Insufficient funds in account ' || p_from_account);  
    ELSE  
        UPDATE Accounts SET Balance = Balance - p_amount WHERE AccountID =  
p_from_account;  
  
        UPDATE Accounts SET Balance = Balance + p_amount WHERE AccountID = p_to_account;  
  
        COMMIT;  
        DBMS_OUTPUT.PUT_LINE('Transfer Successful: $' || p_amount || ' moved from ' ||  
p_from_account || ' to ' || p_to_account);  
    END IF;  
  
EXCEPTION  
    WHEN NO_DATA_FOUND THEN  
        DBMS_OUTPUT.PUT_LINE('Error: One of the Account IDs does not exist.');
```

```
    WHEN OTHERS THEN  
        DBMS_OUTPUT.PUT_LINE('Error: An unexpected error occurred.');
```

```
    ROLLBACK;  
END;  
/
```

Testing Using Some Values:

```
EXEC ProcessMonthlyInterest;
EXEC UpdateEmployeeBonus('Sales', 10);
EXEC TransferFunds(1, 3, 500);
EXEC TransferFunds(2, 3, 10000);
```

Updated Table:

```
BEGIN
```

```
DBMS_OUTPUT.PUT_LINE('-----');
DBMS_OUTPUT.PUT_LINE('          UPDATED ACCOUNTS TABLE');
DBMS_OUTPUT.PUT_LINE('-----');
DBMS_OUTPUT.PUT_LINE('ID   | Name       | Type      | Balance');
DBMS_OUTPUT.PUT_LINE('-----');
```

```
FOR acc IN (SELECT * FROM Accounts ORDER BY AccountID) LOOP
```

```
    DBMS_OUTPUT.PUT_LINE(
        RPAD(acc.AccountID, 5) || ' | ' ||
        RPAD(acc.CustomerName, 15) || ' | ' ||
        RPAD(acc.AccountType, 12) || ' | ' ||
        TO_CHAR(acc.Balance, '99,999.99')
    );
```

```
END LOOP;
```

```
DBMS_OUTPUT.PUT_LINE(CHR(10));
```

```
DBMS_OUTPUT.PUT_LINE('-----');
DBMS_OUTPUT.PUT_LINE('          UPDATED EMPLOYEES TABLE');
DBMS_OUTPUT.PUT_LINE('-----');
DBMS_OUTPUT.PUT_LINE('ID   | Name       | Department | Salary');
DBMS_OUTPUT.PUT_LINE('-----');
```

```
FOR emp IN (SELECT * FROM Employees ORDER BY EmployeeID) LOOP
```

```
    DBMS_OUTPUT.PUT_LINE(
        RPAD(emp.EmployeeID, 5) || ' | ' ||
        RPAD(emp.Name, 15) || ' | ' ||
        RPAD(emp.Department, 12) || ' | ' ||
        TO_CHAR(emp.Salary, '999,999.99')
    );
```

```
END LOOP;
```

```
DBMS_OUTPUT.PUT_LINE('-----');
```

```
END;
```

```
/
```

Output:

Output:
Monthly interest applied to all Savings accounts.
Salaries updated for Department: Sales
Transfer Successful: \$500 moved from 1 to 3
Error: Insufficient funds in account 2

UPDATED ACCOUNTS TABLE

ID	Name	Type	Balance
1	Alice	Savings	4,550.00
2	Bob	Savings	2,020.00
3	Charlie	Checking	1,500.00

UPDATED EMPLOYEES TABLE

ID	Name	Department	Salary
101	John Doe	Sales	55,000.00
102	Jane Smith	Sales	66,000.00
103	Mike Ross	HR	45,000.00
